

Garil AI — University Pilot Budget Pack

GARIL AI · Governed AI for Research, Instruction and Learning · 12-month horizon · USD · September 2026

ANNUAL VPS

\$887.88

AI PER USER

\$4.40

RECOMMENDED PILOT

100 users

YEAR-1 TOTAL @ 100

\$1,327.88

1. Locked commercial inputs

Input	Value
Monthly VPS	\$73.99
Annual VPS	$\$73.99 \times 12 = \887.88
AI cost per user (year seat / allocation)	\$4.40
Recommended first pilot	100 users

VPS specification

Spec	Value
vCPU	8 cores
RAM	32 GB
Disk	400 GB NVMe
Bandwidth	32 TB / month
Monthly price	\$73.99
Annual price	\$887.88

2. Executive summary

Garil AI for a university pilot runs on **one VPS** (application + MongoDB + optional object storage) plus **OpenRouter** for model inference (default main model: GPT-5.1; fast path: gpt-4o-mini).

Item	Figure
Monthly hosting	\$73.99
Annual hosting	\$887.88
AI per user	\$4.40
Recommended cohort	100 users
Year-1 total @ 100 users	\$1,327.88
All-in cost per user @ 100	\$13.28 / year

This VPS is sized comfortably for a **50–500 user** pilot.

3. What runs on the VPS

Service	Role
Next.js frontend	Web UI
Fastify API	Auth, research, portal, admin, WebSocket
MongoDB	Primary database
MinIO (optional)	Large uploads / assets (up to ~2 GB per file when enabled)

Not required for pilot: Redis, Postgres, separate database VPS, or Vercel.

4. Cost model

```

Annual infra    = 73.99 × 12
Annual AI       = users × 4.40
Year-1 total    = 887.88 + (users × 4.40)
Per-user all-in = Year-1 total ÷ users

```

Assumptions

- \$4.40** is the estimated AI/token cost **per active user for one year seat** (full use of a ~1M-token-style allowance on GPT-5.1 blended rates).
- Platform token defaults remain **400,000** (students) and **1,000,000** (lecturers/researchers) unless the university overrides them.
- The in-app admin estimate of **\$2 per 1M tokens** is **not** used here; this pack uses the locked **\$4.40 per user** figure.
- Domain, Resend email, Tavily, and Hugging Face are **optional add-ons** and are excluded from the base totals.

5. Year-1 budget by pilot size

Users	VPS (12 mo)	AI (× \$4.40)	Year-1 total	Per user all-in
50	\$887.88	\$220.00	\$1,107.88	\$22.16
100	\$887.88	\$440.00	\$1,327.88	\$13.28
200	\$887.88	\$880.00	\$1,767.88	\$8.84
500	\$887.88	\$2,200.00	\$3,087.88	\$6.18

Cash-flow view

Line	Cadence	Amount
VPS	Monthly	\$73.99
VPS	Annual	\$887.88
AI budget	Per cohort / year	users × \$4.40 (OpenRouter pay-as-you-go)

With 10% contingency

Users	Base total	Contingency (10%)	Contingency total
50	\$1,107.88	\$110.79	\$1,218.67
100	\$1,327.88	\$132.79	\$1,460.67
200	\$1,767.88	\$176.79	\$1,944.67
500	\$3,087.88	\$308.79	\$3,396.67

6. Recommended pilot design — 100 users

Role	Count	Default tokens	Notes
Students	80	400,000	Primary research / learning users
Lecturers / researchers	15	1,000,000	Supervision & research
Admins	5	n/a	Ops / token management
Total	100	—	AI budget \$440.00 (100 × \$4.40)

For budgeting simplicity, AI is charged at a **flat \$4.40 per user**, not split by role.

Success metrics (Year 1)

- Activation rate of invited users
- Average tokens used vs allowance
- Research papers / supervision reviews completed
- Support tickets and critical incidents

- Actual spend vs budget ($\leq$ **\$1,327.88** base, or $\leq$ **\$1,460.67** with contingency)

7. Included vs not included

Included in this budget	Not included
VPS (\$887.88 / year)	Domain (~\$0–20 / year)
App + Mongo on same server	Resend email (~\$0–20 / mo if heavy)
AI seats at \$4.40 / user	Tavily / Hugging Face (optional)
32 TB bandwidth	Staff time / training
400 GB disk for pilot data	Second VPS / managed MongoDB Atlas

8. Capacity and risk on this VPS

Resource	Pilot assessment
CPU / RAM (8 / 32 GB)	Comfortable for 50–500 lightly concurrent academic users
Disk (400 GB NVMe)	Adequate for pilot; monitor Mongo + MinIO + backups
Bandwidth (32 TB)	Far above typical web/API traffic
Primary cost risk	AI usage if quotas are raised or users exceed plan
Controls	Cap allowances, monitor admin token dashboard, careful resets

9. Go-live checklist

1. Provision the VPS; install Coolify (or Docker Compose).
2. Deploy frontend + backend + MongoDB (+ MinIO if large uploads are required).
3. Configure OPENROUTER_API_KEY, AUTH_SECRET, MONGODB_URI, APP_URL.
4. Set university token defaults (student / lecturer).
5. Invite the pilot cohort; brief admins on the token panel.
6. Track spend monthly: fixed VPS + OpenRouter usage vs AI budget.
7. Review at month 3 and month 12 against Year-1 total **\$1,327.88** (100-user plan).

10. Stakeholder one-pager

Garil AI University Pilot — Year 1

Hosting: 8 vCPU / 32 GB RAM / 400 GB NVMe / 32 TB bandwidth @ **\$73.99/mo (\$887.88/year)**.

AI: **\$4.40 per user**.

Recommended cohort: **100 users** → AI **\$440.00** → **Year-1 total \$1,327.88 (\$13.28 per user)**.

Optional 10% contingency → **\$1,460.67**.

11. Worked example — 100 users

Line	Calculation	Amount (USD)
Annual VPS	73.99×12	887.88
Annual AI	100×4.40	440.00
Year-1 total	$887.88 + 440.00$	1,327.88
Per-user all-in	$1,327.88 \div 100$	13.28
With 10% contingency	$1,327.88 \times 1.10$	1,460.67

Prepared for university pilot planning. Figures are locked commercial assumptions for this document; actual OpenRouter invoices will vary with real token consumption.